



ST BENEDICT'S COLLEGE

SUBJECT	LIFE SCIENCES	DATE	16 MAY 2022
GRADE	11	MARKS	50
EXAMINER	MRS S. STEGMANN	MODERATOR	MRS B. ZAJAC
NAME		DURATION	50 MINUTES
CLASS			

INSTRUCTIONS:

1. This paper consists of 6 pages. Please check that your question paper is complete.
 2. Answer all questions on the folio paper provided.
 3. Write in blue or black pen only.
 4. Read all questions carefully and answer in as much detail as possible.
 5. All diagrams should have a clear heading.
 6. It is in your own interest to write legibly and to present your work neatly.
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QUESTION	POSSIBLE MARK	MARK ACHIEVED
1	10	
2	15	
3	15	
4	10	
TOTAL: 50		

Question 1:

1.1. Various possibilities are suggested as answers to the following questions. Indicate the correct answer by writing the correct letter from the corresponding question on the folio paper provided.

1.1.1. Which statement is incorrect regarding the heart?

- A. Its walls are made up of striated voluntary muscle
- B. It is situated in a space between the lungs called the mediastinum
- C. Internally it is divided into four chambers
- D. The upper atria are receiving chambers with relatively thin walls

1.1.2. Mammals are said to have a “double circulation” because...

- A. blood makes two full circuits of the body every five minutes.
- B. There are two blood circuits from the heart: one to the lungs and the other to the rest of the body.
- C. Each organ is served by two vessels: an artery and a vein.
- D. The heart has two valves, thus providing double the circulatory power of other hearts.

1.1.3. The following events occur during a heartbeat.

- i. Semilunar valves forced open
- ii. Blood flows into atria
- iii. Both ventricles contract
- iv. Mitral and tricuspid valves open

Select the correct order in which these events occur:

- A. i, ii, iii, iv
- B. iii, iv, i, ii
- C. ii, iv, iii, i
- D. ii, iv, i, iii

1.1.4. Which of the following statements is correctly paired with a part of the circulatory system?

Part of the circulatory system	Statement
A. Pulmonary vein	Carries oxygenated blood at low pressure
B. Semilunar valve	Prevents flow of blood from arteries into ventricles
C. AV valves	Prevents flow of blood from arteries into ventricles
D. Aorta	Carries deoxygenated blood at high pressure

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1.1.5. Assuming a healthy heart, if a person has a heart rate of 72 beats per minute, calculate the duration it takes to complete each cardiac cycle.

- A. 1,2 seconds
- B. 1,2 minutes
- C. 0,8 seconds
- D. 0,8 minutes

(5)

1.2. Provide the correct biological term for the following statements

1.2.1. The only veins to transport oxygenated blood.

1.2.2. Fluid that is drained between the cells of the tissue and transported back to the bloodstream.

1.2.3. The relaxation of the cardiac muscle.

1.2.4. White blood cells that actively leave capillaries to engulf foreign substances.

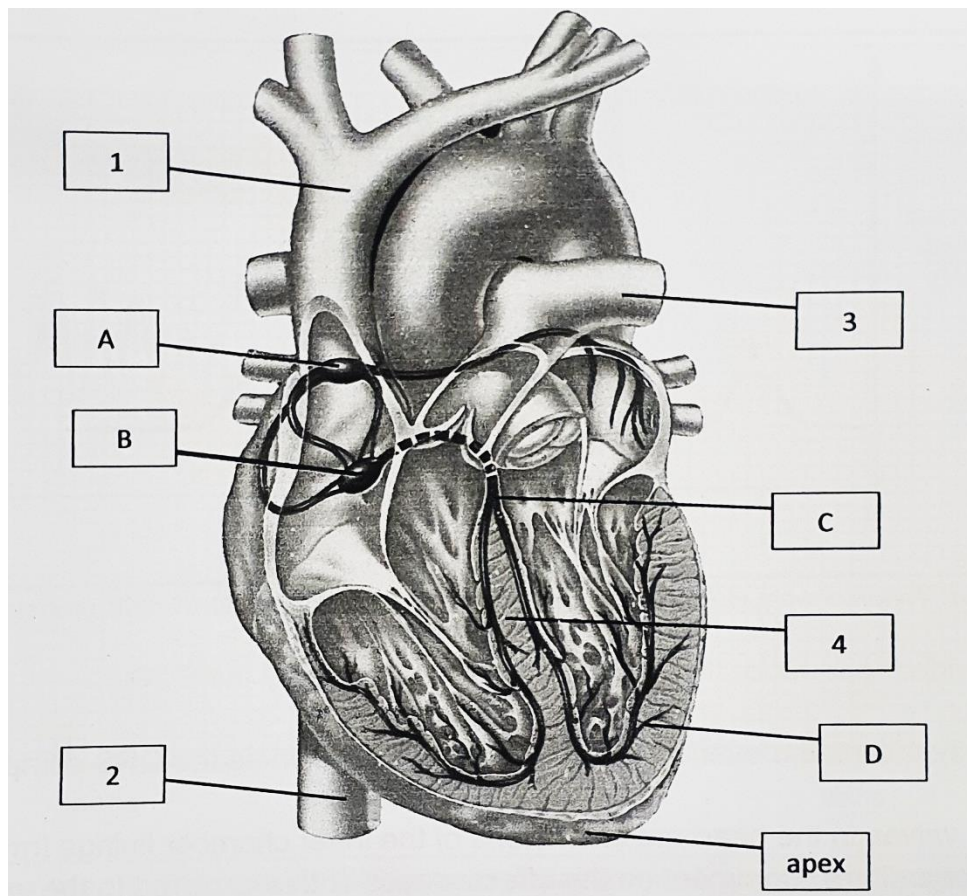
1.2.5. The number of heart beats per minute.

(5)

[10]

Question 2:

Study the diagram of the heart below and answer the questions that follow.

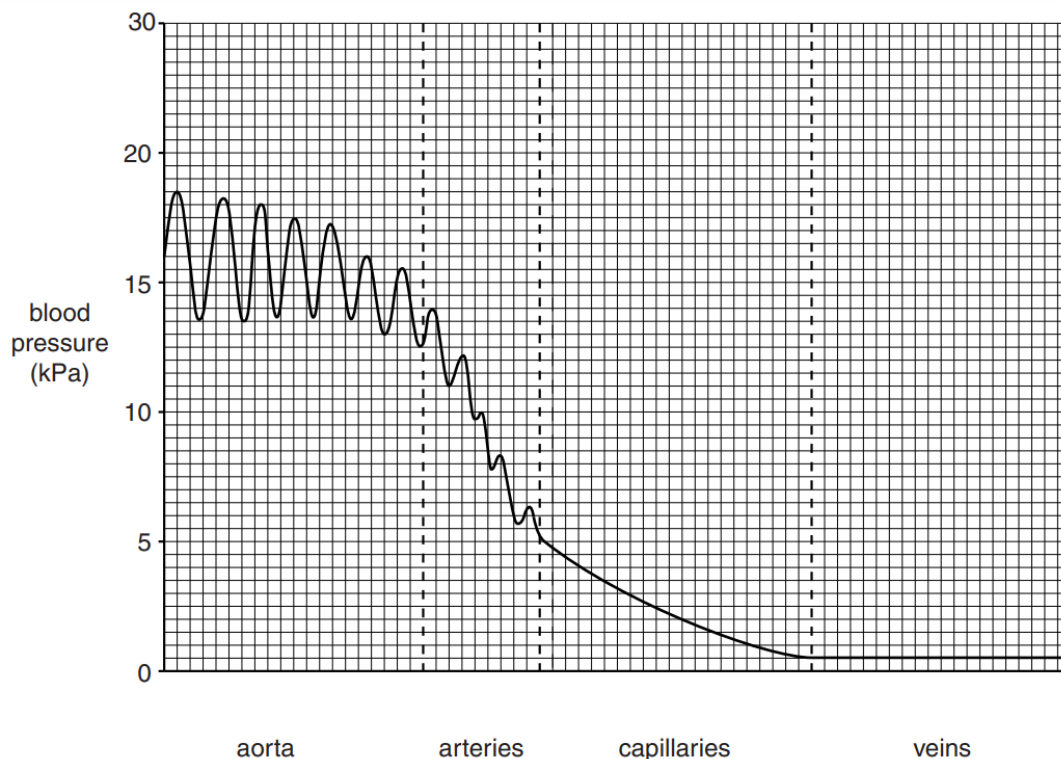


- 2.1. Provide labels for parts numbered 1, 2 and 3. (3)
- 2.2. Explain the importance of structure 4. (2)
- 2.3. Structures A, B, C and D are involved in conducting the heartbeat.
- 2.3.1. The heart muscle is myogenic. Define this term. (1)
- 2.3.2. Provide the letter and the name of the structure which generates the heartbeat. (2)
- 2.3.3. By using the **names** of structures A-D, explain how the heartbeat is generated. (4)
- 2.3.4. Although the structure mentioned in 2.3.2. initiates the heartbeat, it cannot alter the rate at which the heartbeat contracts. What system is involved in slowing the heart rate and how does it achieve this? (3)

[15]

Question 3

The graph below provides information about the blood pressure in different parts of the mammalian blood circulatory system.



- 3.1. What is the highest blood pressure achieved in the arteries? (2)

- 3.2. Explain why a higher blood pressure is achieved and maintained in the aorta. (2)
- 3.3. The pressure fluctuates as the blood flows along the aorta and arteries. (2)
- a. What causes this fluctuation? (2)
 - b. Draw a labelled diagram of a cross section of an artery to indicate the various layers that make up this vessel. (5)
- 3.4. The capillaries are the smallest blood vessel (with extremely small lumens) yet the pressure is drastically reduced in capillary networks. (1)
- a. How is this drop in pressure achieved? (1)
 - b. Why is it necessary that the pressure changes as blood flows into the capillaries? (2)
- 3.5. Although veins operate under the lowest pressure, blood is still effectively transported in these vessels. What structural adaptation do these vessels contain to aid the movement of blood back towards the heart? (1)

[15]

Question 4

Blood pressure refers to the two pressures measured in the arteries when the muscles of the heart contract and relax. The instrument that is used to measure blood pressure is known as a sphygmomanometer around the upper arm, as pictured below.

A nurse at a wellness centre assessed 3 different patients and recorded the following respective readings: 135/85 mmHG; 120/80 mmHG and 140/90 mmHG.



- 4.1. Construct a table to record the nurse's data. (4)
- 4.2. Of the three patients, which reading would be in a normal, healthy range? (1)
- 4.3. Consider the reading of 140/90 mmHG. The value: **140** represents which phase of the cardiac cycle? (1)
- 4.4. Provide the term for chronic high blood pressure. (1)
- 4.5. State one factor that cannot be controlled and one factor that can be controlled when it comes to high blood pressure in an individual. (2)
- 4.6. Name one danger associated with abnormally high blood pressure. (1)

[10]

TOTAL: 50 MARKS